

Video pipeline benchmark (new)

Real Y4M→H.264→native CAVLC embed→blind extraction→strict decode runs. Higher resolutions are scaled from CIF and must not be interpreted as native high-resolution scenes.

Source	Resolution	Sample fr	Source fr	Full sec	Encode ms	Capacity ms	Embed ms	CPU s	FPS*	RSS MB	Gate
akiyo_cif.y4m	352x288	30	300	10.0	74.8	1508.2	86.3	0.062	347.7	9.1	passed
akiyo_cif.y4m	640x480	30	300	10.0	116.7	6157.0	269.3	0.297	111.4	16.2	passed
akiyo_cif.y4m	1280x960	30	300	10.0	260.3	27586.2	1070.5	1.047	28.0	40.4	passed
city_cif.y4m	352x288	30	300	10.0	85.4	3397.2	163.7	0.188	183.2	10.9	passed
city_cif.y4m	640x480	30	300	10.0	139.7	14351.1	580.2	0.547	51.7	19.6	passed
city_cif.y4m	1280x960	30	300	10.0	331.5	85964.1	3254.5	3.172	9.2	52.3	passed
coastguard_cif.y4m	352x288	30	300	10.0	82.3	2973.9	151.2	0.109	198.5	10.7	passed
coastguard_cif.y4m	640x480	30	300	10.0	133.6	13865.5	564.0	0.547	53.2	19.5	passed
coastguard_cif.y4m	1280x960	30	300	10.0	332.2	91771.5	3195.7	3.172	9.4	52.3	passed
container_cif.y4m	352x288	30	300	10.0	81.2	2733.1	139.4	0.125	215.2	10.2	passed
container_cif.y4m	640x480	30	300	10.0	134.1	11011.0	448.0	0.438	67.0	17.9	passed
container_cif.y4m	1280x960	30	300	10.0	308.8	63847.1	2344.3	2.312	12.8	48.1	passed
deadline_cif.y4m	352x288	30	1374	45.8	83.9	3043.0	147.8	0.125	203.0	10.4	passed
deadline_cif.y4m	640x480	30	1374	45.8	136.1	11770.8	480.0	0.453	62.5	18.3	passed
deadline_cif.y4m	1280x960	30	1374	45.8	312.3	64476.3	2317.3	2.312	12.9	48.1	passed
football_cif.y4m	352x288	30	260	8.7	85.8	3817.9	178.8	0.172	167.8	11.2	passed
football_cif.y4m	640x480	30	260	8.7	143.2	15805.3	677.0	0.672	44.3	20.5	passed
football_cif.y4m	1280x960	30	260	8.7	350.3	96250.7	3868.5	3.844	7.8	54.7	passed
foreman_cif.y4m	352x288	30	50	5.0	69.0	582.1	42.9	0.031	699.2	7.6	passed
foreman_cif.y4m	640x480	30	50	5.0	95.2	2199.7	95.9	0.078	312.9	12.0	passed

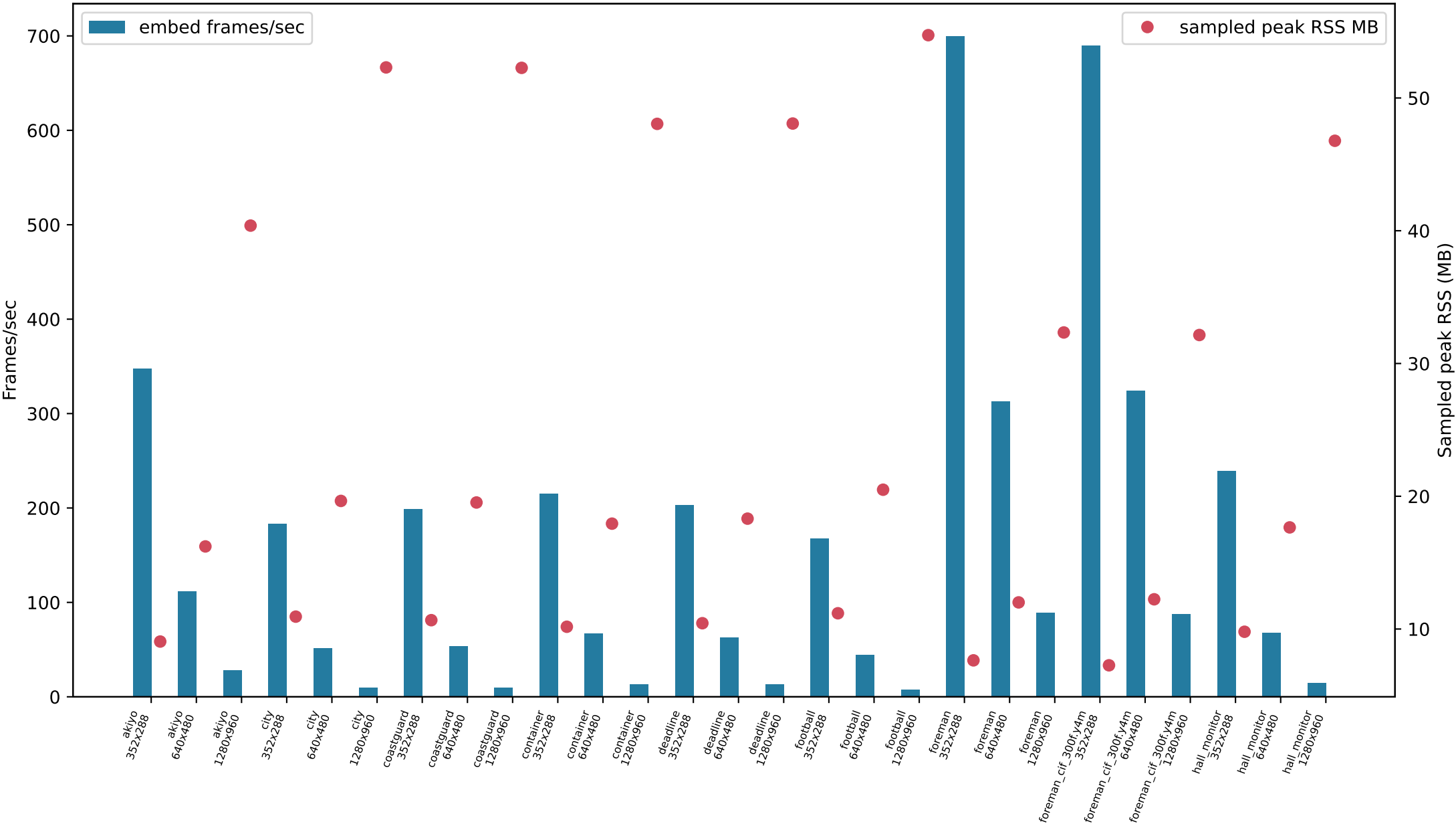
Video pipeline benchmark (new) — continued

Real Y4M→H.264→native CAVLC embed→blind extraction→strict decode runs. Higher resolutions are scaled from CIF and must not be interpreted as native high-resolution scenes.

Source	Resolution	Sample fr	Source fr	Full sec	Encode ms	Capacity ms	Embed ms	CPU s	FPS*	RSS MB	Gate
foreman_cif.y4m	1280x960	30	50	5.0	194.2	9593.6	336.9	0.328	89.0	32.3	passed
foreman_cif_300f.y4m	352x288	30	300	30.0	66.7	578.3	43.5	0.031	689.6	7.3	passed
foreman_cif_300f.y4m	640x480	30	300	30.0	96.7	2191.1	92.6	0.094	323.8	12.2	passed
foreman_cif_300f.y4m	1280x960	30	300	30.0	203.0	9613.0	342.0	0.328	87.7	32.1	passed
hall_monitor_cif.y4m	352x288	30	300	10.0	81.4	2508.6	125.7	0.109	238.7	9.8	passed
hall_monitor_cif.y4m	640x480	30	300	10.0	131.3	11114.6	444.1	0.422	67.6	17.7	passed
hall_monitor_cif.y4m	1280x960	30	300	10.0	312.0	56304.7	2059.5	2.062	14.6	46.8	passed

Embedding performance and resource use

*Output FPS = encoded frames / native embed wall time; not sensor/camera acquisition FPS. RSS and CPU are sampled only for the native embed child.



Per-frame visual quality summary

Raw-source comparison includes H.264 compression plus embedding. Cover-to-stego isolates embedding impact. Full per-frame PSNR/SSIM measurements are in quality_per_frame_new.csv.

Source	Size	Frames	Source PSNR μ	PSNR min	Source SSIM μ	SSIM min	Embed PSNR min	Embed SSIM min
akiyo_cif.y4m	352x288	30	45.025	34.930	0.984862	0.950511	35.290	0.961177
akiyo_cif.y4m	640x480	30	46.317	33.510	0.987366	0.970163	33.680	0.980659
akiyo_cif.y4m	1280x960	30	48.282	40.810	0.990906	0.985761	41.550	0.993854
city_cif.y4m	352x288	30	43.089	38.900	0.986952	0.968763	40.850	0.979294
city_cif.y4m	640x480	30	44.421	37.830	0.989248	0.980642	38.870	0.990382
city_cif.y4m	1280x960	30	45.948	39.680	0.990852	0.987291	40.780	0.995957
coastguard_cif.y4m	352x288	30	42.848	37.140	0.987767	0.967239	38.400	0.977570
coastguard_cif.y4m	640x480	30	44.230	36.710	0.990017	0.979378	37.510	0.988349
coastguard_cif.y4m	1280x960	30	45.826	40.250	0.991254	0.987123	41.550	0.995537
container_cif.y4m	352x288	30	43.631	37.480	0.982994	0.952800	38.550	0.965819
container_cif.y4m	640x480	30	45.002	36.220	0.987409	0.973982	36.800	0.985115
container_cif.y4m	1280x960	30	46.716	40.830	0.990648	0.985831	42.020	0.994525
deadline_cif.y4m	352x288	30	43.770	35.670	0.987687	0.964626	36.310	0.973731
deadline_cif.y4m	640x480	30	45.169	39.220	0.989250	0.976537	40.380	0.985691
deadline_cif.y4m	1280x960	30	46.807	41.990	0.991250	0.986563	43.580	0.994567
football_cif.y4m	352x288	30	43.242	37.520	0.987327	0.970482	38.870	0.981025
football_cif.y4m	640x480	30	44.578	39.010	0.988966	0.981176	40.470	0.991175
football_cif.y4m	1280x960	30	46.195	41.880	0.990275	0.987232	43.970	0.996410
foreman_cif.y4m	352x288	30	53.349	41.800	0.998894	0.994378	42.100	0.995080
foreman_cif.y4m	640x480	30	51.818	40.990	0.997761	0.988720	41.300	0.990022
foreman_cif.y4m	1280x960	30	53.414	39.880	0.998132	0.992937	40.040	0.994210
foreman_cif_300f.y4m	352x288	30	53.392	43.070	0.998949	0.996039	43.430	0.996723
foreman_cif_300f.y4m	640x480	30	51.758	39.180	0.997753	0.988477	39.420	0.989822
foreman_cif_300f.y4m	1280x960	30	53.410	39.770	0.998129	0.992828	39.900	0.994037
hall_monitor_cif.y4m	352x288	30	43.861	35.510	0.978761	0.945736	36.020	0.961932
hall_monitor_cif.y4m	640x480	30	45.556	40.290	0.984355	0.970475	41.690	0.984258
hall_monitor_cif.y4m	1280x960	30	47.062	32.650	0.989044	0.982073	32.760	0.991990

Scope and acceptance

Benchmark status and reproducibility boundaries.

Item	Value
Pass gate	native embed + strict FFmpeg decode + blind correct-key recovery + wrong-key rejection
Source corpus	akiyo_cif.y4m, city_cif.y4m, coastguard_cif.y4m, container_cif.y4m, deadline_cif.y4m, foo
Encoding	FFmpeg libx264, Baseline/CAVLC, QP 22, all-intra, one thread and one slice per IDR; this
Quality metric	Per-frame FFmpeg PSNR Y and SSIM measured both from source Y4M to stego decode (to
ZKP scope	Native authenticated CAVLC stream benchmark embeds HMAC-protected payloads. Grot
Run count	27 passed / 27 total
Run ID	20260924T150707Z_2be60c

Full-duration Y4M to H.264 conversion (new)

All source frames are encoded at each Y4M file's native resolution. This is distinct from the bounded 30-frame, multi-resolution embedding matrix.

Source	Resolution	Frames	Duration s	Encode ms	H.264 bytes	Output FPS	Gate
akiyo_cif.y4m	352x288	300	10.01	287.5	4769837	1043.6	converted
city_cif.y4m	352x288	300	10.00	362.0	10040988	828.8	converted
coastguard_cif.y4m	352x288	300	10.01	353.2	10571442	849.3	converted
container_cif.y4m	352x288	300	10.01	330.4	8392432	907.9	converted
deadline_cif.y4m	352x288	1374	45.85	1425.8	43120070	963.7	converted
football_cif.y4m	352x288	260	8.67	314.6	7796840	826.4	converted
foreman_cif.y4m	352x288	50	5.00	74.8	567523	668.5	converted
foreman_cif_300f.y4m	352x288	300	30.00	201.6	3402313	1487.9	converted
hall_monitor_cif.y4m	352x288	300	10.01	330.8	7060727	906.9	converted